Using the reduction formulas I and II, we obtain the following result (stated without proofs):

$$\int_0^{\frac{\pi}{2}} \sin^n x\, dx = \int_0^{\frac{\pi}{2}} \cos^n x\, dx = \begin{cases} \frac{(n-1)}{n} \times \frac{(n-3)}{(n-2)} \times \cdots \times \frac{1}{2} \times \frac{\pi}{2}, & \text{if } n = 2,4,6,\cdots \\ \frac{(n-1)}{n} \times \frac{(n-3)}{(n-2)} \times \cdots \times \frac{2}{3}, & \text{if } n = 3,5,7,\cdots \end{cases}$$

Note As illustrations, we have

$$\int_0^{\frac{\pi}{2}} \cos^5 x\, dx = \int_0^{\frac{\pi}{2}} \sin^5 x\, dx = \frac{4}{5} \times \frac{2}{3} \times 1$$

$$\int_0^{\frac{\pi}{2}} \sin^6 x\, dx = \int_0^{\frac{\pi}{2}} \cos^6 x\, dx = \frac{5}{6} \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2}$$

Example 9.37

Evaluate $\int_0^{\frac{\pi}{2}} \left(\sin^2 x + \cos^4 x\right) dx$

Solution

Given that $I = \int_0^{\frac{\pi}{2}} \left(\sin^2 x + \cos^4 x\right) dx = \int_0^{\frac{\pi}{2}} \sin^2 x\, dx + \int_0^{\frac{\pi}{2}} \cos^4 x\, dx = \frac{1}{2} \times \frac{\pi}{2} + \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} = \frac{7\pi}{16}$.

Example 9.38

Evaluate $\int_0^{\frac{\pi}{2}} \begin{vmatrix} \cos^4 x & 7 \\ \sin^5 x & 3 \end{vmatrix} dx$.

Solution

$$I = \int_0^{\frac{\pi}{2}} \left(3\cos^4 x - 7\sin^5 x\right) dx = 3\int_0^{\frac{\pi}{2}} \cos^4 x\, dx - 7\int_0^{\frac{\pi}{2}} \sin^5 x\, dx$$

$$= 3 \times \frac{3}{4} \times \frac{1}{2} \times \frac{\pi}{2} - 7 \times \frac{4}{5} \times \frac{2}{3} = \frac{9\pi}{16} - \frac{56}{15}.$$

By applying the reduction formula III iteratively, we get the following results (stated without proof):

(i) If n is even and m is even,

$$\int_0^{\frac{\pi}{2}} \sin^m x \cos^n x\, dx = \frac{(n-1)}{(m+n)} \frac{(n-3)}{(m+n-2)} \frac{(n-5)}{(m+n-4)} \cdots \frac{1}{(m+2)} \frac{(m-1)}{m} \frac{(m-3)}{(m-2)} \frac{(m-5)}{(m-4)} \cdots \frac{1}{2} \frac{\pi}{2}$$

(ii) If n is odd and m is any positive integer (even or odd), then

$$\int_0^{\frac{\pi}{2}} \sin^m x \cos^n x\, dx = \frac{(n-1)}{(m+n)} \frac{(n-3)}{(m+n-2)} \frac{(n-5)}{(m+n-4)} \cdots \frac{2}{(m+3)} \frac{1}{(m+1)}.$$

Note

If one of m and n is odd, then it is convenient to get the power of $\cos x$ as odd. For instance, if m is odd and n is even, then

$$\int_0^{\frac{\pi}{2}} \sin^m x \cos^n x\, dx = \int_0^{\frac{\pi}{2}} \sin^n x \cos^m x\, dx = \frac{(m-1)}{(n+m)} \frac{(m-3)}{(n+m-2)} \frac{(m-5)}{(n+m-4)} \cdots \frac{2}{(n+3)} \frac{1}{(n+1)}.$$

Example 9.39

Find the values of the following:

(i) $\int_0^{\frac{\pi}{2}} \sin^5 x \cos^4 x\, dx$ (ii) $\int_0^{\frac{\pi}{2}} \sin^4 x \cos^6 x\, dx$

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